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Analysis of factors associated with injury severity in freeway crashes based on XGBoost–SHAP algorithm

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Introduction

- High injury risk and socioeconomic losses associated with high-speed expressway crashes
- Difficulty in identifying injury-risk conditions due to context-dependent crash outcomes
- Limitations of prior injury-severity studies assuming linear and additive effects
- Need for nonlinear, interaction-aware modeling under severe class imbalance
- Use of 2011–2019 nationwide expressway crash data with binary injury occurrence outcome
- Development of an interpretable XGBoost-based injury prediction model
- SHAP-based identification of key injury-risk factors and effect directions
- Support for proactive, data-driven expressway safety management

Research Objective

Identification of injury-occurrence factors in expressway crashes and interpretation of their effect directions using SHAP, providing an interpretable basis for proactive safety management and countermeasure prioritization

Methodology

01 Data & preprocessing
- Use of 2011–2019 nationwide expressway crash records
- Binary definition of injury occurrence outcome
- Removal of post-crash response and outcome variables
- One-hot encoding of categorical variables into 9,684 predictors

02 Model Development
- XGBoost binary classifier
- scale_pos_weight = 8.26 for class imbalance
- F1-based threshold optimization

03 Evaluation & SHAP
- Evaluation metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC, PR-AUC
- Model interpretation using gain importance and SHAP analysis
- SHAP-based identification of risk-increasing/decreasing factors

Data description and Outcome Variable

Item	Value
Source	Nationwide expressway crash data from 2011 to 2019
Target	Injury occurrence (binary classification)
Records	83,297 records (Training: 66,637 / Test: 16,660)
Class balance	Injury cases: 10.8%; imbalance ratio: approx. 8.3:1
Predictors	9,684 pre-accident predictors after one-hot encoding
Leakage control	Post-crash response and outcome variables removed

Pre-accident Predictor variables

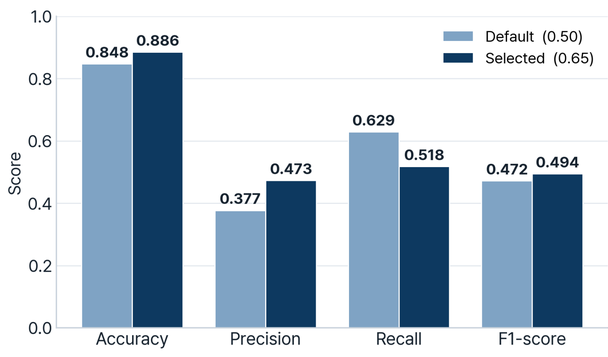
Category	Examples
Crash	Crash type, No. of vehicles, main cause, pre-crash maneuver
Driver / Vehicle	Driver state, sex, age group, vehicle type & size
Road geometry	Route, milepost, lane, alignment, superelevation
Environment / Location	Weather, road surface, day-night, hour, branch, mainline / tunnel / tollgate

Model Framework

Item	Value
Model type	XGBoost binary classification model
Feature scope	Only pre-accident information was used
Imbalance handling	Class-weight adjustment to emphasize minority injury cases
Threshold optimization	The classification threshold was tuned to maximize F1-score
Data split	Stratified 80:20 train–test split
Performance evaluation	Accuracy, Precision, Recall, F1-score, ROC-AUC, and PR-AUC

Model Evaluation and Risk Factor Interpretation

1. Model performance

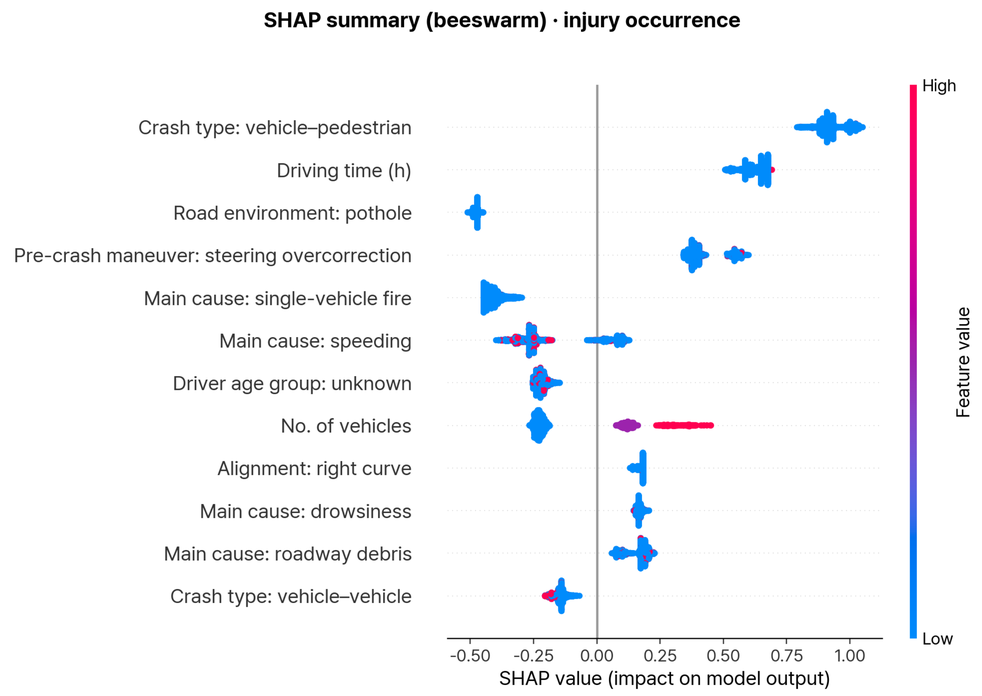


Metric	Default	Selcted
Accuracy	0.848	**0.886**
Precision	0.377	**0.473**
Recall	0.629	0.518
F1-score	0.472	**0.494**
ROC-AUC	0.853	0.853
PR-AUC	0.499	0.499

Selected threshold = 0.65 (best F1)

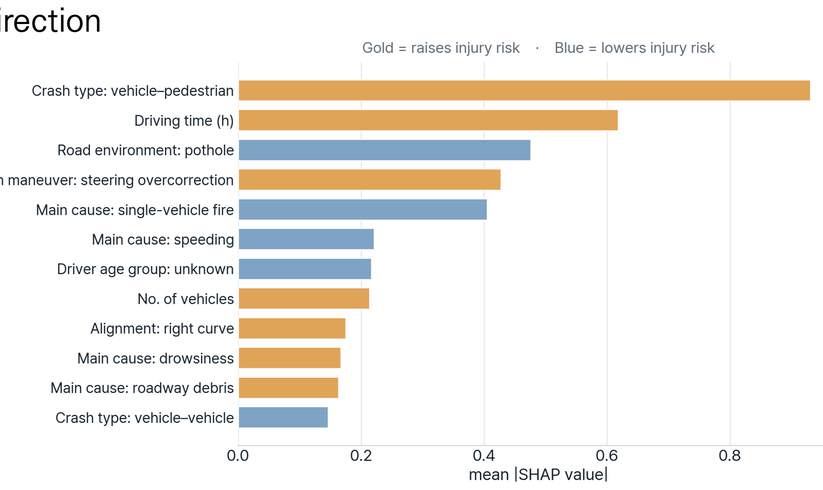
- Limited reliability of Accuracy under severe class imbalance
- Use of Precision, Recall, F1-score, ROC-AUC, and PR-AUC for balanced performance evaluation
- Strong discrimination performance based on ROC-AUC = 0.853
- Meaningful injury-risk signal under imbalance, with PR-AUC = 0.499 above the prevalence baseline
- Improvement of Precision and F1-score through threshold optimization from 0.50 to 0.65
- Precision–Recall trade-off observed after threshold adjustment
- Selection of threshold 0.65 as the best F1 operating point

2. SHAP summary–injury-risk factors



- Dominant injury-risk contribution of crash dynamics
- Strong risk increase from vehicle–pedestrian crashes and steering overcorrection
- Driver fatigue effects from long driving time and drowsiness
- Increased injury risk associated with multiple vehicles and roadway debris
- Lower predicted injury risk in pothole environments, single-vehicle fires, and vehicle–vehicle crashes
- Dominant contribution of crash-cause variables to injury-risk prediction

3. Risk direction



- Risk increase: vehicle–pedestrian crashes, long driving time, steering overcorrection, drowsiness, roadway debris
- Risk decrease: potholes, single-vehicle fire, vehicle–vehicle crashes
- Bar length: average impact based on mean |SHAP|
- Counter-intuitive signs treated as exploratory findings

Conclusion & future work

- Potential of interpretable machine learning for expressway injury occurrence prediction
- Moderate but improvable predictive performance under severe class imbalance ROC-AUC = 0.853, PR-AUC = 0.499
- Improvement of Precision and F1-score through threshold optimization
- Identification of key injury-risk contributors using SHAP
- Major risk factors: vehicle–pedestrian crashes, long driving time, steering overcorrection, drowsiness, and roadway debris
- Future improvement through additional external data, refined predictors, and stronger imbalance-handling methods

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